

Research Lab Equipment Booking & Tracking System

Advanced Experiment Lifecycle Management

Team Name: DEIMOS

Team Members:

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Technologies:

HTML, CSS, JavaScript, PHP, MySQL

Presented To: Golam Hossain

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Introduction

The Challenge

Manual lab booking systems are inefficient, leading to scheduling conflicts and poor equipment usage tracking. Communication gaps often delay research.

The Solution

Our digital platform centralizes all laboratory operations, ensuring transparency, accountability, and real-time coordination for all researchers.

System Features (Overview)

- Equipment booking system
- Experiment tracking
- Role-based access control
- Chat system
- Damage reporting
- Report generation (PDF)

System Stakeholders



Admin



Supervisors



Students



Guests

Admin Features

- Manage users (approve/reject)
- Add/remove supervisors
- Manage equipment categories
- View reports
- Manage safety policies

Supervisor Features

- Approve/reject bookings
- Update equipment status
- Track experiments
- Upload SOP & safety instructions
- Report damages

Researcher & Student Portal

Researchers

Submit experiment plans, book high-value tools, and upload results directly.

Students

Access supervised bookings, maintain lab notes, and view real-time schedules.

Public Guest Access



Information Visibility

Guests can view lab descriptions, equipment availability, and essential safety protocols without system credentials.

System Stakeholders



Admin



Supervisors



Researchers



Students



Guests

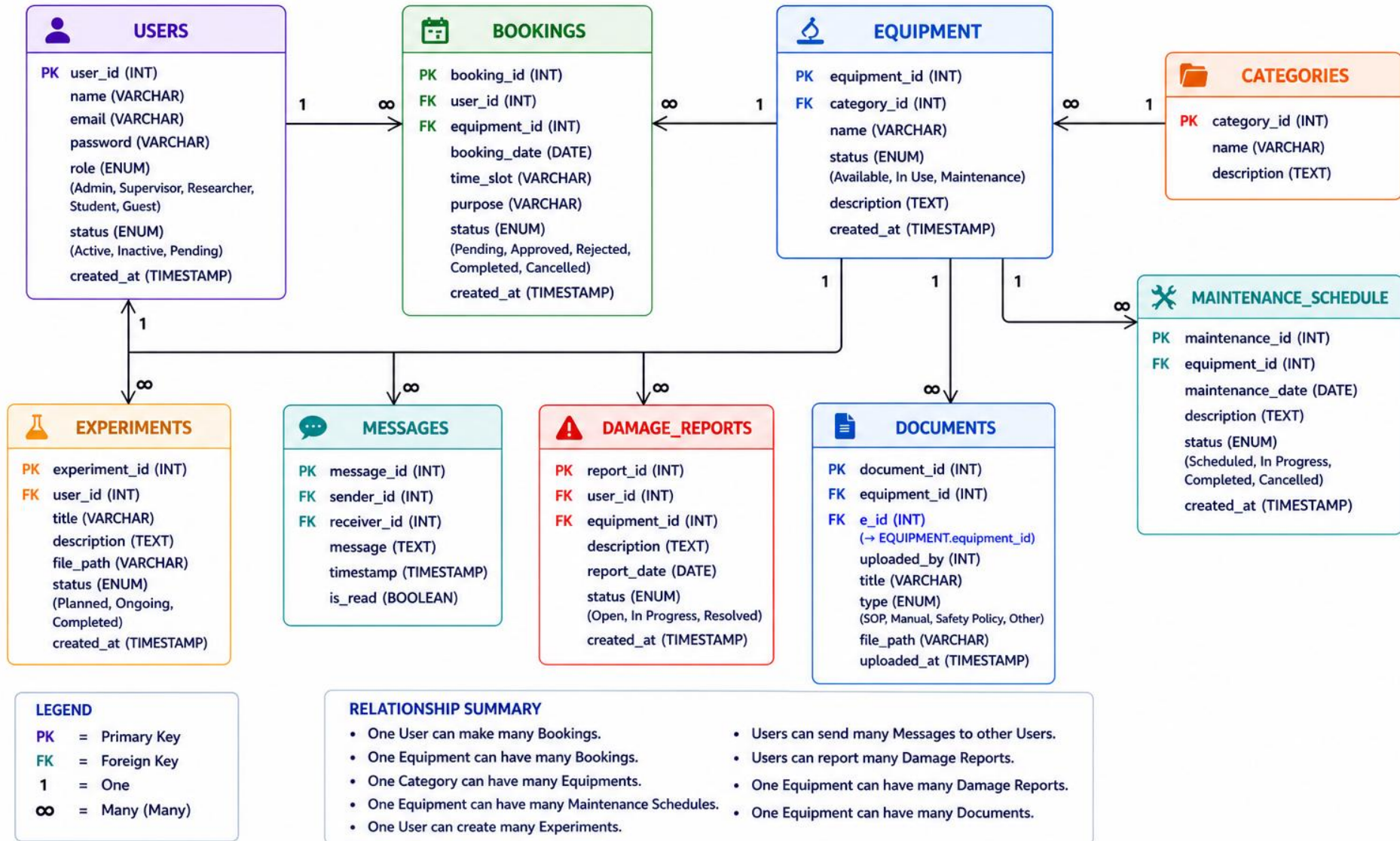


System Architecture

A 3-tier architecture ensuring data security, processing efficiency, and an intuitive user interface.

Research Lab Equipment Booking & Experiment Tracking System

• ERD (Entity Relationship Diagram) •



Frontend Work Distribution

UI Development

- Member 1: Authentication UI
- Member 2: Dashboard Design
- Member 3: Booking Pages

Interaction

- Member 4: File Upload Modals
- Member 5: Chat Interface

Backend Work Distribution

Server Logic

- Member 1: Auth Backend
- Member 2: Booking Algorithms
- Member 3: MySQL Connection

Data Services

- Member 4: File Stream Handling
- Member 5: Chat & API Backend

System Workflow Lifecycle



Implementation Challenges

-  Complex role-based permission inheritance
-  Securing large research file uploads
-  Real-time conflict resolution for equipment
-  Database normalization and scaling

Future Improvements

Alerts

Push & Email Notifications

Analytics

Advanced Usage Dashboard

Mobile

iOS & Android App version

Conclusion

Our system transforms lab management from a manual burden into a streamlined digital asset, fostering innovation through better resource tracking and communication.

Thank You!

Any Questions?